General Mathematics (Further Maths 2020–22 set) · Easier paper

Real VCAA questions ranked by the published state results. Editable: type your answers straight into this PDF, save it, then mark yourself with the guide at the back.

Name _____ **Date** _____

Exam 1 · multiple choice	Exam 2 · written response	State average on these questions
40 marks	**61 marks**	**74 / 101 (73%)**
reading 15 min · writing 1 h 30 min	reading 15 min · writing 1 h 30 min	a student at the state average on every question

How to use this paper

1. Open each question with its blue link. It opens the official VCAA exam PDF at the right page. The question text isn't reproduced here because it is VCAA copyright.
2. Answer in this file: click a bubble for multiple choice, and type in the boxes for written answers (long answers scroll inside the box). Working can be typed in plain text, e.g. 0.004 × 5440.88 = 21.76.
3. Keep to the real time limits above, with your CAS, your bound reference and the VCAA formula sheet, and don't look at the guide until you've finished.
4. Save with Ctrl/Cmd + S. Use a PDF app that supports forms: Adobe Acrobat Reader, Chrome, Edge, Firefox, macOS Preview or Xodo.
5. Mark yourself: in the guide at the back, tick each marking point you earned, enter your marks, and compare your total with the state average.

Why these questions

These are real VCAA questions from the 2020–2022 Further Mathematics exams, the subject General Maths replaced in 2023. Only the core (data analysis, recursion and finance) and the Matrices and Networks modules are used, and questions on content that isn't in the current General Maths study design (solving simultaneous equations with matrices) are left out. They were ranked by how Further Maths students actually scored on them within each area of study, and this paper takes the highest band.

Sources

- 2020: Exam 1 (PDF) · report · Exam 2 (PDF) · report
- 2021: Exam 1 (PDF) · report · Exam 2 (PDF) · report
- 2022: Exam 1 (PDF) · report · Exam 2 (PDF) · report

Exam 1 · multiple choice (40 marks, reading 15 min, writing 1 h 30 min)

Open each question in the official exam PDF, then click your answer. Further Maths questions have five options (A–E).

Data analysis

Question 1 · 1 mark

2020 VCAA Further Maths Exam 1 · Core · Question 1 · open in the official exam PDF (p. 4) ↗

Difference between the mean and the median of 800 nerve-impulse times, from a summary table

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 2 · 1 mark

2020 VCAA Further Maths Exam 1 · Core · Question 2 · open in the official exam PDF (p. 5) ↗

How many of the 800 times are longer than 300 ms

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 3 · 1 mark

2020 VCAA Further Maths Exam 1 · Core · Question 5 · open in the official exam PDF (p. 6) ↗

Percentage of 20 animal species lighter than 10 000 g, from a $\log_{10}$(weight) histogram

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 4 · 1 mark

2020 VCAA Further Maths Exam 1 · Core · Question 6 · open in the official exam PDF (p. 6) ↗

Which variable a percentage segmented bar chart could pair with month of the year

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 5 · 1 mark

2020 VCAA Further Maths Exam 1 · Core · Question 8 · open in the official exam PDF (p. 7) ↗

Number of 10 000 birds with wing length below 57 mm (mean 61 mm, SD 2 mm)

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 6 · 1 mark

2020 VCAA Further Maths Exam 1 · Core · Question 9 · open in the official exam PDF (p. 7) ↗

Mean and SD of light-globe lifetimes from two given percentages

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 7 · 1 mark

2020 VCAA Further Maths Exam 1 · Core · Question 11 · open in the official exam PDF (p. 8) ↗

Percentage of study participants who did not snore, from a two-way table

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 8 · 1 mark

2020 VCAA Further Maths Exam 1 · Core · Question 14 · open in the official exam PDF (p. 9) ↗

Predicted number of tasks after 3 hours from a $\log_{10}$(tasks) model

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 9 · 1 mark

2020 VCAA Further Maths Exam 1 · Core · Question 17 · open in the official exam PDF (p. 11) ↗

Deseasonalised rainfall for May 2019

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 10 · 1 mark

2020 VCAA Further Maths Exam 1 · Core · Question 18 · open in the official exam PDF (p. 11) ↗

Six-mean smoothed rainfall with centring for August 2019

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 11 · 1 mark

2021 VCAA Further Maths Exam 1 · Core · Question 2 · open in the official exam PDF (p. 5) ↗

Which statement about a percentage segmented bar chart supports an association between destination and age

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 12 · 1 mark

2022 VCAA Further Maths Exam 1 · Core · Question 2 · open in the official exam PDF (p. 5) ↗

Percentage of 46 possums with skull width below 55 mm

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 13 · 1 mark

2022 VCAA Further Maths Exam 1 · Core · Question 4 · open in the official exam PDF (p. 5) ↗

Mean and standard deviation of 14 possum ages from a dot plot

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 14 · 1 mark

2022 VCAA Further Maths Exam 1 · Core · Question 5 · open in the official exam PDF (p. 6) ↗

Number of 2498 possums with body length between 84 and 96 cm (mean 88, SD 4)

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 15 · 1 mark

2022 VCAA Further Maths Exam 1 · Core · Question 6 · open in the official exam PDF (p. 6) ↗

Number of seals with spleen weight of at least 1000 g, from a $\log_{10}$ histogram

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 16 · 1 mark

2022 VCAA Further Maths Exam 1 · Core · Question 11 · open in the official exam PDF (p. 9) ↗

Two-way column-percentage table of qualified (A or B grade) against instructional method

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Recursion and financial modelling

Question 17 · 1 mark

2020 VCAA Further Maths Exam 1 · Core · Question 21 · open in the official exam PDF (p. 13) ↗

Which term of $T_0 = 10$, $T_{n+1} = T_n + 3$ equals 13

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 18 · 1 mark

2020 VCAA Further Maths Exam 1 · Core · Question 23 · open in the official exam PDF (p. 13) ↗

How many of four recurrence relations show a depreciating asset

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 19 · 1 mark

2020 VCAA Further Maths Exam 1 · Core · Question 24 · open in the official exam PDF (p. 14) ↗

Interest earned in the first five months from $B_0 = 3000$, $B_{n+1} = 1.0048B_n$

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 20 · 1 mark

2020 VCAA Further Maths Exam 1 · Core · Question 25 · open in the official exam PDF (p. 14) ↗

Recurrence relation matching a graph of an annuity balance that stays level

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 21 · 1 mark

2020 VCAA Further Maths Exam 1 · Core · Question 27 · open in the official exam PDF (p. 15) ↗

Years for \$10 000 at 5.5% p.a. compounding annually to first exceed double

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 22 · 1 mark

2021 VCAA Further Maths Exam 1 · Core · Question 17 · open in the official exam PDF (p. 14) ↗

Constant C in $L_{n+1} = L_n + C$, given $L_0 = 37$ and $L_2 = 25$

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 23 · 1 mark

2021 VCAA Further Maths Exam 1 · Core · Question 18 · open in the official exam PDF (p. 14) ↗

Principal reduction for payment 3 from an annuity amortisation table

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 24 · 1 mark

2022 VCAA Further Maths Exam 1 · Core · Question 18 · open in the official exam PDF (p. 12) ↗

Month when the balance from $V_{n+1} = 1.003V_n - 2024$ first falls below \$398 000

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Matrices

Question 25 · 1 mark

2020 VCAA Further Maths Exam 1 · Matrices module · Question 1 · open in the official exam PDF (p. 17) ↗

Type of a given 3 × 3 matrix of 0s and 1s

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 26 · 1 mark

2020 VCAA Further Maths Exam 1 · Matrices module · Question 2 · open in the official exam PDF (p. 17) ↗

Calculation that gives element q_{41} of a matrix product Q = AB

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 27 · 1 mark

2020 VCAA Further Maths Exam 1 · Matrices module · Question 4 · open in the official exam PDF (p. 19) ↗

Transition matrix matching a transition diagram for three magazines

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 28 · 1 mark

2020 VCAA Further Maths Exam 1 · Matrices module · Question 5 · open in the official exam PDF (p. 20) ↗

Communication matrix matching a diagram of direct links between four people

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 29 · 1 mark

2020 VCAA Further Maths Exam 1 · Matrices module · Question 6 · open in the official exam PDF (p. 21) ↗

Build a 3 × 3 matrix from the rule $m_{ij} = 3i + 2j$

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 30 · 1 mark

2021 VCAA Further Maths Exam 1 · Matrices module · Question 1 · open in the official exam PDF (p. 18) ↗

Transpose of a 3 × 2 matrix

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 31 · 1 mark

2022 VCAA Further Maths Exam 1 · Matrices module · Question 1 · open in the official exam PDF (p. 16) ↗

Element holding the daily cost of an adult mountain bike

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 32 · 1 mark

2022 VCAA Further Maths Exam 1 · Matrices module · Question 4 · open in the official exam PDF (p. 18) ↗

Sequence of links that lets Ursula reach Steph, from a communication matrix

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Networks and decision mathematics

Question 33 · 1 mark

2020 VCAA Further Maths Exam 1 · Networks module · Question 1 · open in the official exam PDF (p. 24) ↗

Number of faces of a connected planar graph with seven vertices and nine edges

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 34 · 1 mark

2020 VCAA Further Maths Exam 1 · Networks module · Question 2 · open in the official exam PDF (p. 24) ↗

Which listed route is not a Hamiltonian cycle

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 35 · 1 mark

2020 VCAA Further Maths Exam 1 · Networks module · Question 3 · open in the official exam PDF (p. 25) ↗

Which of five trees is not a spanning tree of a given network

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 36 · 1 mark

2020 VCAA Further Maths Exam 1 · Networks module · Question 8 · open in the official exam PDF (p. 28) ↗

Which pathway network matches an adjacency matrix (including a loop)

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 37 · 1 mark

2021 VCAA Further Maths Exam 1 · Networks module · Question 1 · open in the official exam PDF (p. 22) ↗

Number of vertices of degree 3 in a graph

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 38 · 1 mark

2021 VCAA Further Maths Exam 1 · Networks module · Question 2 · open in the official exam PDF (p. 22) ↗

Which statement about a bipartite graph of friends and the fruit they ate is false

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 39 · 1 mark

2021 VCAA Further Maths Exam 1 · Networks module · Question 4 · open in the official exam PDF (p. 23) ↗

Number of vertices of a directed network that cannot be reached from X

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 40 · 1 mark

2022 VCAA Further Maths Exam 1 · Networks module · Question 1 · open in the official exam PDF (p. 21) ↗

What Prim’s algorithm finds

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Exam 2 · written response (61 marks, reading 15 min, writing 1 h 30 min)

Open each question in the official exam PDF, then type your answer to every part in the boxes. Show working where more than one mark is available.

Data analysis

Question 1 (3 marks)

2020 VCAA Further Maths Exam 2 · Core · Question 1 · open in the official exam PDF (p. 2) ↗

An ordered stem plot of the BMI of 32 men: shape, median and the percentage overweight

a. Describe the shape of the distribution (1 mark)

b. Median BMI (1 mark)

c. Percentage of the men with a BMI of 25 or more (1 mark)

Question 2 (5 marks)

2020 VCAA Further Maths Exam 2 · Core · Question 4 · open in the official exam PDF (p. 8) ↗

Age, body density and weight of 12 men: summary statistics and a least squares line for body density against weight

a.i. Median age (1 mark)

a.ii. Mean body density (1 mark)

b.i. Explanatory variable when predicting body density from weight (1 mark)

b.ii. Slope of the least squares line, to three significant figures (1 mark)

c. Percentage of variation in body density explained by weight (1 mark)

Question 3 (3 marks)

2021 VCAA Further Maths Exam 2 · Core · Question 2 · open in the official exam PDF (p. 5) ↗

A least squares line relating 800 m and 200 m running times, with each variable's mean and standard deviation

a. Round the intercept and slope to three significant figures (1 mark)

b. Coefficient of determination as a percentage, from the slope and standard deviations (2 marks)

Question 4 (5 marks)

2022 VCAA Further Maths Exam 2 · Core · Question 2 · open in the official exam PDF (p. 4) ↗

Parallel boxplots of relative humidity at 9 am and 3 pm over 30 days

a.i. Is humidity more variable at 9 am or 3 pm? Quote both IQRs (1 mark)

a.ii. Percentage of days with 3 pm humidity below 80% (1 mark)

a.iii. Complete the five-number summaries (1 mark)

b. Do the boxplots support an association between humidity and time of day? Quote a statistic (2 marks)

Question 5 (4 marks)

2022 VCAA Further Maths Exam 2 · Core · Question 3 · open in the official exam PDF (p. 6) ↗

Eight days of weather data: variable types, median rainfall, and a least squares line for maximum against minimum temperature

a.i. Number of numerical variables in the table (1 mark)

a.ii. Median rainfall (1 mark)

b. Round the intercept and slope to four significant figures (1 mark)

c. Coefficient of determination as a percentage, to one decimal place (1 mark)

Question 6 (5 marks)

2022 VCAA Further Maths Exam 2 · Core · Question 4 · open in the official exam PDF (p. 8) ↗

Relative humidity against temperature at 9 am over 30 days, with a least squares line ($r^2 = 0.6073$)

a. Explanatory variable (1 mark)

b. Describe the association: strength and direction (1 mark)

c. Interpret the slope (1 mark)

d. Residual for the day with temperature 16.3 °C and humidity 45.0%, to one decimal place (2 marks)

Recursion and financial modelling

Question 7 (4 marks)

2020 VCAA Further Maths Exam 2 · Core · Question 7 · open in the official exam PDF (p. 14) ↗

A printing machine depreciated by flat rate: $V_0 = 120\,000$, $V_{n+1} = V_n - 15\,000$

a. Yearly decrease in value (1 mark)

b. Value after two years, showing recursive calculations (1 mark)

c. Annual flat rate percentage (1 mark)

d. Rule of the form $V_n = a + bn$ (1 mark)

Question 8 (4 marks)

2022 VCAA Further Maths Exam 2 · Core · Question 6 · open in the official exam PDF (p. 12) ↗

Equipment depreciated by flat rate: $V_n = 200\,000 - 12\,500n$

a. Value after one year (1 mark)

b. Years until the value first reaches zero (1 mark)

c. Equivalent recurrence relation (1 mark)

d. Name for depreciating by a fixed percentage of the current value each year (1 mark)

Question 9 (4 marks)

2022 VCAA Further Maths Exam 2 · Core · Question 7 · open in the official exam PDF (p. 13) ↗

Amortisation table for a $540 000 annuity at 3% p.a. paying $5214.28 a month for 10 years

a. Value of payment number 3 (1 mark)

b. Interest for payment 3, to the nearest cent (1 mark)

c. Recurrence relation for the balance P_n (1 mark)

d. Monthly payment if the $540 000 were a simple perpetuity (1 mark)

Matrices

Question 10 (4 marks)

2020 VCAA Further Maths Exam 2 · Matrices module · Question 3 · open in the official exam PDF (p. 23) ↗

Shoppers moving week to week between three centres after one is sold: a transition matrix and a graph of the next 10 weeks

a. State matrix one week after the sale (1 mark)

b. Change in Westmall's weekly shoppers from the sale to week 10, to the nearest thousand (1 mark)

c. Week in which Grandmall reaches its maximum (1 mark)

d. Long-term weekly number of shoppers at Westmall, to the nearest whole number (1 mark)

Question 11 (2 marks)

2021 VCAA Further Maths Exam 2 · Matrices module · Question 1 · open in the official exam PDF (p. 16) ↗

Bottles of three olive oil brands sold in January, stored as a column matrix

a. Order of matrix J (1 mark)

b. Scalar k that increases every element by 5% (1 mark)

Question 12 (3 marks)

2021 VCAA Further Maths Exam 2 · Matrices module · Question 2 · open in the official exam PDF (p. 16) ↗

A communication matrix for five office staff and its square

a. Two staff who can send information directly to each other (1 mark)

b. Sequence of links from Elena to Chai (1 mark)

c. List both two-step links for the pair with two different two-step links (1 mark)

Question 13 (2 marks)

2022 VCAA Further Maths Exam 2 · Matrices module · Question 1 · open in the official exam PDF (p. 16) ↗

Nightly costs of three types of ski-resort accommodation stored as a row matrix

a. Matrix equation for the cost of five nights of each type (1 mark)

b. Column matrix A so that CA gives the cost of 2 motel nights and 3 apartment nights (1 mark)

Networks and decision mathematics

Question 14 (3 marks)

2020 VCAA Further Maths Exam 2 · Networks module · Question 1 · open in the official exam PDF (p. 26) ↗

A graph showing which of five new cricketers had played together before

a. How many players Emerson had played with (1 mark)

b. Who had played with both Alex and Bo (1 mark)

c. Add a new player, Finn, who had played with none of them (1 mark)

Question 15 (1 mark)

2020 VCAA Further Maths Exam 2 · Networks module · Question 2 · open in the official exam PDF (p. 27) ↗

Assigning three players to batting positions 1–3 to maximise their total average score

Batting position for each player that maximises the team's score (1 mark)

Question 16 (4 marks)

2021 VCAA Further Maths Exam 2 · Networks module · Question 1 · open in the official exam PDF (p. 22) ↗

A house floor plan with five rooms and the outside, drawn as a graph with one edge missing

a. Draw the missing edge (1 mark)

b. Degree of vertex E (1 mark)

c.i. A route from F through every room once and back to F (1 mark)

c.ii. Mathematical name for such a journey (1 mark)

Question 17 (2 marks)

2021 VCAA Further Maths Exam 2 · Networks module · Question 2 · open in the official exam PDF (p. 23) ↗

A network of roads between towns G and M

a. Shortest distance from G to M (1 mark)

b. Town passed through twice on the shortest route that visits every town (1 mark)

Question 18 (3 marks)

2022 VCAA Further Maths Exam 2 · Networks module · Question 1 · open in the official exam PDF (p. 22) ↗

A network of roads joining an office and seven holiday cabins

a. A route that visits every cabin once and returns to the office (1 mark)

b.i. Vertex where a route along every road exactly once, starting at the office, finishes (1 mark)

b.ii. Name for that route (1 mark)

Marking guide

Mark yourself only after you've finished both exams. Tick the boxes you earned, enter your marks and compare with how Victoria went on each question.

Exam 1 · multiple choice

✓	Q	Ans	Why	How Victoria went
☐	1	**B**	Mean 220 − median 150 = 70 ms.	98% correct A 1% · B 98% · C 1% · D 0% · E 0%
☐	2	**D**	300 ms is the third quartile, so a quarter of the values lie above it: 0.25 × 800 = 200.	73% correct A 3% · B 6% · C 12% · D 73% · E 7%
☐	3	**E**	10 000 = 10^4, so count the bars below 4 on the log scale: 3 + 2 + 12 = 17 of 20 species, which is 85%.	78% correct A 8% · B 4% · C 5% · D 4% · E 78%
☐	4	**D**	A segmented bar chart shows two categorical variables. Rainfall recorded as below average/average/above average is the only categorical option.	72% correct A 17% · B 7% · C 1% · D 72% · E 3%
☐	5	**D**	57 is two standard deviations below the mean, so 2.5% lie below it: 0.025 × 10 000 = 250.	78% correct A 3% · B 6% · C 8% · D 78% · E 4%
☐	6	**D**	50% above 670 means the mean is 670. 16% below 655 means 655 is one SD below the mean, so SD = 15.	77% correct A 2% · B 8% · C 9% · D 77% · E 3%
☐	7	**E**	91 ÷ 161 = 56.5%.	95% correct A 0% · B 1% · C 2% · D 1% · E 95%
☐	8	**C**	$\log_{10}$(tasks) = 1.160 + 0.03617 × 3 = 1.2685, so tasks = 10^1.2685 ≈ 18.6, closest to 19.	74% correct A 14% · B 6% · C 74% · D 4% · E 2%
☐	9	**B**	92.6 ÷ 1.222 ≈ 75.8 mm.	88% correct A 2% · B 88% · C 4% · D 3% · E 3%
☐	10	**C**	Mean of May–October is 80.93 and of June–November is 81.72; centring averages them: 81.3 mm.	73% correct A 5% · B 8% · C 73% · D 9% · E 4%
☐	11	**D**	An association means the percentages differ between the age groups: 65% of under-55s prefer domestic compared with 45% of those 55 and over. A single percentage can't show it.	82% correct A 1% · B 5% · C 2% · D 82% · E 10%
☐	12	**B**	Add the bars below 55: 1 + 1 + 2 + 1 + 2 + 5 = 12, and 12 ÷ 46 ≈ 26%.	85% correct A 7% · B 85% · C 7% · D 1% · E 0%
☐	13	**E**	Enter the 14 ages into CAS: mean ≈ 4.93 and sample SD ≈ 2.53, so 4.9 and 2.5.	74% correct A 4% · B 3% · C 5% · D 14% · E 74%
☐	14	**A**	84 is one SD below and 96 two SDs above: 34% + 47.5% = 81.5%, and 0.815 × 2498 ≈ 2036.	76% correct A 76% · B 6% · C 8% · D 7% · E 3%

☐	15	**D**	1000 g = 10^3, so add the bars from 3.0 up: 8 + 9 + 7 + 1 = 25.	75% correct A 5% · B 7% · C 10% · D 75% · E 2%
☐	16	**B**	Individual: (10 + 35) ÷ 115 ≈ 39% qualified; group: (18 + 30) ÷ 126 ≈ 38%. Only B has 39/61 and 38/62.	86% correct A 4% · B 86% · C 6% · D 2% · E 2%
☐	17	**A**	$T_1 = 10 + 3 = 13$.	85% correct A 85% · B 5% · C 2% · D 1% · E 7%
☐	18	**C**	$V_n - 2500$ and $0.875V_n$ both fall. $1.125V_n - 2500$ stays at 20 000 (interest 2500 cancels the 2500 taken off), and + 2500 grows. Two.	80% correct A 1% · B 5% · C 80% · D 14% · E 1%
☐	19	**D**	$3000 \times 1.0048^5 - 3000 \approx \$72.69$. C used simple interest.	80% correct A 3% · B 8% · C 5% · D 80% · E 3%
☐	20	**B**	A level balance means the interest each period equals the payment. Only B works: 0.025 × 200 000 = 5000.	79% correct A 8% · B 79% · C 5% · D 3% · E 4%
☐	21	**B**	$1.055^{12} \approx 1.90$ and $1.055^{13} \approx 2.006$, so 13 years.	77% correct A 6% · B 77% · C 6% · D 5% · E 5%
☐	22	**A**	Two steps change the value by 2C: 37 + 2C = 25, so C = −6.	83% correct A 83% · B 5% · C 4% · D 6% · E 2%
☐	23	**C**	Payment − interest = 44 970.55 − 17 962.40 = \$27 008.15.	93% correct A 1% · B 3% · C 93% · D 3% · E 1%
☐	24	**C**	Iterate: $V_1 \approx 399\,176$, $V_2 \approx 398\,349.53$, $V_3 \approx 397\,520.58$. Month 3.	78% correct A 8% · B 9% · C 78% · D 3% · E 1%
☐	25	**A**	Every element is 0 or 1, so it is a binary matrix. It isn't symmetric, triangular or a permutation matrix (row 2 has two 1s).	83% correct A 83% · B 10% · C 1% · D 2% · E 4%
☐	26	**E**	Row 4 of A times column 1 of B: 4 × 2 + 5 × 4.	79% correct A 1% · B 4% · C 7% · D 8% · E 79%
☐	27	**C**	Each column is "this week" and must add to 1. Reading the arrows out of A, B and C gives C. D wrongly adds a 35% flow from A to C.	90% correct A 1% · B 2% · C 90% · D 5% · E 2%
☐	28	**D**	Put a 1 in row sender, column receiver for each arrow (two-way arrows give two 1s). Only D matches every arrow; E wrongly uses 2s.	86% correct A 4% · B 4% · C 3% · D 86% · E 3%
☐	29	**B**	Row 1: 5, 7, 9; row 2: 8, 10, 12; row 3: 11, 13, 15.	73% correct A 7% · B 73% · C 8% · D 7% · E 4%
☐	30	**D**	Rows become columns: the first row of M^T is 3, 8, 13 and the second is 2, 9, 7.	82% correct A 2% · B 10% · C 2% · D 82% · E 4%
☐	31	**E**	Adult is row 3 and mountain bike is column 2: b_{32}.	83% correct A 2% · B 1% · C 8% · D 6% · E 83%

☐	32	**C**	Read rows as senders: U can send to T, T to W, and W to S. U-T-S fails because T can't send to S.	87% correct A 5% · B 2% · C 87% · D 4% · E 1%
☐	33	**D**	Euler's formula: 7 − 9 + f = 2, so f = 4.	89% correct A 1% · B 2% · C 5% · D 89% · E 3%
☐	34	**D**	DCBAGFED needs an edge from E to D, which the graph doesn't have. The others visit every vertex once along real edges.	93% correct A 1% · B 2% · C 2% · D 93% · E 2%
☐	35	**B**	Option B uses an edge from 3 to 5, which isn't in the network. The others use only network edges, reach all six vertices and have five edges.	81% correct A 5% · B 81% · C 8% · D 3% · E 3%
☐	36	**E**	Check each entry: a loop at J, 3 paths J–K, 2 J–M, 1 K–L, 2 K–M and 2 L–M. Only E has them all.	75% correct A 3% · B 9% · C 8% · D 5% · E 75%
☐	37	**E**	Count the edges at each vertex: five vertices have degree 3, one has degree 5 and one has degree 2.	91% correct A 1% · B 1% · C 1% · D 5% · E 91%
☐	38	**C**	Van's vertex has two edges (orange and strawberry), so "Van ate only strawberry" is false.	95% correct A 1% · B 2% · C 95% · D 1% · E 1%
☐	39	**A**	Follow the arrows from X: it reaches W, then T, V, U, Y and Z. S only has edges leaving it, so S is the one vertex that can't be reached.	62% correct A 62% · B 8% · C 3% · D 7% · E 20%
☐	40	**E**	A minimum spanning tree.	80% correct A 6% · B 9% · C 3% · D 2% · E 80%

Multiple-choice score: ☐ **/ 40** **(state average 32.7)**

Exam 2 · written response

Question 1 (3 marks) · state average 2.4 / 3 (80%)

2020 VCAA Further Maths Exam 2 · Core · Question 1

a. Describe the shape of the distribution

☐ Positively skewed

Answer / notes: Positively skewed: most values sit between 21 and 25 and they tail off up to 31. Adding "no outliers" was accepted.

State average 0.9 / 1 (90%); 86% scored full marks. Mark distribution: 0: 14% · 1: 86%

Your mark: ☐ **/ 1**

b. Median BMI

☐ 24.55

Answer / notes: 32 values, so the median is the mean of the 16th and 17th: (24.5 + 24.6) ÷ 2 = 24.55. Many gave 24.5 or 24.6.

State average 0.7 / 1 (70%); 70% scored full marks. Mark distribution: 0: 30% · 1: 70%

Your mark: ☐ **/ 1**

c. Percentage of the men with a BMI of 25 or more

☐ 37.5%

Answer / notes: 12 of the 32 men: 37.5%. Rounding to 38% lost the mark.

State average 0.8 / 1 (80%); 84% scored full marks. Mark distribution: 0: 16% · 1: 84%

Your mark: ☐ **/ 1**

Question 2 (5 marks) · state average 3.4 / 5 (68%)

2020 VCAA Further Maths Exam 2 · Core · Question 4

a.i. Median age

☐ 24

Answer / notes: 24 years.

State average 1.0 / 1 (100%); 97% scored full marks. Mark distribution: 0: 3% · 1: 97%

Your mark: ☐ **/ 1**

a.ii. Mean body density

☐ 1.065

Answer / notes: 12.78 ÷ 12 = 1.065 kg/L. Don't round when no rounding is asked for.

State average 0.8 / 1 (80%); 77% scored full marks. Mark distribution: 0: 23% · 1: 77%

Your mark: ☐ **/ 1**

b.i. Explanatory variable when predicting body density from weight

☐ weight

Answer / notes: Weight.

State average 0.8 / 1 (80%); 78% scored full marks. Mark distribution: 0: 22% · 1: 78%

Your mark: ☐ **/ 1**

b.ii. Slope of the least squares line, to three significant figures

☐ −0.00112

Answer / notes: CAS gives −0.0011156…, so −0.00112. Many lost the negative sign or rounded wrongly.

State average 0.3 / 1 (30%); 29% scored full marks. Mark distribution: 0: 71% · 1: 29%

Your mark: ☐ **/ 1**

c. Percentage of variation in body density explained by weight

☐ 29%

Answer / notes: $r^2 \approx 0.290$, so 29%.

State average 0.5 / 1 (50%); 54% scored full marks. Mark distribution: 0: 46% · 1: 54%

Your mark: ☐ **/ 1**

Question 3 (3 marks) · state average 1.1 / 3 (37%)

2021 VCAA Further Maths Exam 2 · Core · Question 2

a. Round the intercept and slope to three significant figures

☐ 0.0393 and 5.28

Answer / notes: time800 = 0.0393 + 5.28 × time200. Many rounded to three decimal places instead.

State average 0.5 / 1 (50%); 52% scored full marks. Mark distribution: 0: 48% · 1: 52%

Your mark: ☐ **/ 1**

b. Coefficient of determination as a percentage, from the slope and standard deviations

- [ ] $r = 5.2756 \times 0.96956 \div 8.2910 \approx 0.617$
- [ ] $r^2 \approx 38\%$

Answer / notes: $b = r \times s_y \div s_x$, so $r = b \times s_x \div s_y \approx 0.617$ and $r^2 \approx 0.381$: 38%.

State average 0.6 / 2 (30%); 29% scored full marks. Mark distribution: 0: 67% · 1: 3% · 2: 29%

Your mark: ☐ **/ 2**

Question 4 (5 marks) · state average 3.3 / 5 (66%)

2022 VCAA Further Maths Exam 2 · Core · Question 2

a.i. Is humidity more variable at 9 am or 3 pm? Quote both IQRs

- [ ] 9 am, because its IQR (35) is larger than 3 pm's (29)

Answer / notes: IQR at 9 am ≈ 100 − 65 = 35 and at 3 pm ≈ 80 − 51 = 29, so 9 am is more variable. You had to state the conclusion as well as the values.

State average 0.6 / 1 (60%); 59% scored full marks. Mark distribution: 0: 42% · 1: 59%

Your mark: ☐ **/ 1**

a.ii. Percentage of days with 3 pm humidity below 80%

- [ ] 75%

Answer / notes: 80% is the 3 pm Q3, so 75% of days are below it.

State average 0.7 / 1 (70%); 70% scored full marks. Mark distribution: 0: 30% · 1: 70%

Your mark: ☐ **/ 1**

a.iii. Complete the five-number summaries

- [ ] 9 am: minimum 40, Q3 100, maximum 100; 3 pm: Q3 80, maximum 100

Answer / notes: Read the missing values off the boxplots. At 9 am Q3 and the maximum are both 100.

State average 0.9 / 1 (90%); 90% scored full marks. Mark distribution: 0: 10% · 1: 90%

Your mark: ☐ **/ 1**

b. Do the boxplots support an association between humidity and time of day? Quote a statistic

- [ ] Yes: the median changes with time of day
- [ ] Median 75% at 9 am compared with 63% at 3 pm

Answer / notes: Yes. The median humidity is 75% at 9 am but 63% at 3 pm. Quoting the medians without saying they differ lost a mark.

State average 1.1 / 2 (55%); 41% scored full marks. Mark distribution: 0: 36% · 1: 22% · 2: 41%

Your mark: ☐ **/ 2**

Question 5 (4 marks) · state average 2.4 / 4 (60%)

2022 VCAA Further Maths Exam 2 · Core · Question 3

a.i. Number of numerical variables in the table

- [ ] 5

Answer / notes: Minimum and maximum temperature, rainfall, maximum wind speed and 9 am temperature. Day number is a label and wind direction is categorical. 6 counted day number.

State average 0.6 / 1 (60%); 58% scored full marks. Mark distribution: 0: 42% · 1: 58%

Your mark: ☐ **/ 1**

a.ii. Median rainfall

- [ ] 0 mm

Answer / notes: Five of the eight days had 0 mm, so the median is 0. Some found the mean.

State average 0.7 / 1 (70%); 70% scored full marks. Mark distribution: 0: 30% · 1: 70%

Your mark: ____ **/ 1**

b. Round the intercept and slope to four significant figures

- [ ] 9.236 and 1.002

Answer / notes: maximum temperature = 9.236 + 1.002 × minimum temperature. The slope was often rounded to four decimal places instead.

State average 0.6 / 1 (60%); 62% scored full marks. Mark distribution: 0: 38% · 1: 62%

Your mark: ____ **/ 1**

c. Coefficient of determination as a percentage, to one decimal place

- [ ] 61.1%

Answer / notes: CAS gives $r^2 \approx 0.6112$, so 61.1%. 0.6 wasn't a percentage.

State average 0.5 / 1 (50%); 48% scored full marks. Mark distribution: 0: 52% · 1: 48%

Your mark: ____ **/ 1**

Question 6 (5 marks) · state average 3.1 / 5 (62%)

2022 VCAA Further Maths Exam 2 · Core · Question 4

a. Explanatory variable

- [ ] temperature

Answer / notes: Temperature.

State average 1.0 / 1 (100%); 93% scored full marks. Mark distribution: 0: 7% · 1: 93%

Your mark: ____ **/ 1**

b. Describe the association: strength and direction

- [ ] strong, negative

Answer / notes: $r = -\sqrt{0.6073} \approx -0.78$, which is strong and negative. "Moderate" came from using r^2 instead of r.

State average 0.3 / 1 (30%); 28% scored full marks. Mark distribution: 0: 72% · 1: 28%

Your mark: ____ **/ 1**

c. Interpret the slope

- [ ] On average, relative humidity decreases by 3.417% for each 1 °C increase in temperature

Answer / notes: Humidity decreases by 3.417 percentage points, on average, for each 1 °C increase. Leaving out "each 1 degree" lost the mark.

State average 0.5 / 1 (50%); 44% scored full marks. Mark distribution: 0: 56% · 1: 44%

Your mark: ____ **/ 1**

d. Residual for the day with temperature 16.3 °C and humidity 45.0%, to one decimal place

- [ ] Predicted = 120.1 − 3.417 × 16.3 = 64.40
- [ ] Residual = 45.0 − 64.40 = −19.4

Answer / notes: Residual = actual − predicted = −19.4. Many stopped at the predicted value.

State average 1.3 / 2 (65%); 54% scored full marks. Mark distribution: 0: 23% · 1: 23% · 2: 54%

Your mark: ____ **/ 2**

Question 7 (4 marks) · state average 2.7 / 4 (68%)

2020 VCAA Further Maths Exam 2 · Core · Question 7

a. Yearly decrease in value

☐ $15 000

Answer / notes: $15 000.

State average 0.9 / 1 (90%); 94% scored full marks. Mark distribution: 0: 6% · 1: 94%

Your mark: ____ **/ 1**

b. Value after two years, showing recursive calculations

☐ $V_1 = 120\,000 - 15\,000 = 105\,000$ and $V_2 = 105\,000 - 15\,000 = 90\,000$

Answer / notes: $90 000, with both steps shown. Some didn't use recursion as asked.

State average 0.7 / 1 (70%); 72% scored full marks. Mark distribution: 0: 28% · 1: 72%

Your mark: ____ **/ 1**

c. Annual flat rate percentage

☐ 12.5%

Answer / notes: 15 000 ÷ 120 000 = 12.5%.

State average 0.7 / 1 (70%); 75% scored full marks. Mark distribution: 0: 25% · 1: 75%

Your mark: ____ **/ 1**

d. Rule of the form $V_n = a + bn$

☐ $V_n = 120\,000 - 15\,000n$

Answer / notes: $V_n = 120\,000 - 15\,000n$. Some wrote the recurrence relation again instead of a rule.

State average 0.4 / 1 (40%); 44% scored full marks. Mark distribution: 0: 56% · 1: 44%

Your mark: ____ **/ 1**

Question 8 (4 marks) · state average 3.1 / 4 (78%)

2022 VCAA Further Maths Exam 2 · Core · Question 6

a. Value after one year

☐ $187 500

Answer / notes: 200 000 − 12 500 = $187 500.

State average 1.0 / 1 (100%); 94% scored full marks. Mark distribution: 0: 6% · 1: 94%

Your mark: ____ **/ 1**

b. Years until the value first reaches zero

☐ 16

Answer / notes: 200 000 ÷ 12 500 = 16 years.

State average 1.0 / 1 (100%); 88% scored full marks. Mark distribution: 0: 12% · 1: 88%

Your mark: ____ **/ 1**

c. Equivalent recurrence relation

☐ $V_0 = 200\,000$, $V_{n+1} = V_n - 12\,500$

Answer / notes: $V_0 = 200\,000$, $V_{n+1} = V_n - 12\,500$. Writing "− 12 500n" at the end was a common error.

State average 0.6 / 1 (60%); 62% scored full marks. Mark distribution: 0: 38% · 1: 62%

Your mark: ____ **/ 1**

d. Name for depreciating by a fixed percentage of the current value each year

☐ Reducing balance depreciation

Answer / notes: Reducing balance.

State average 0.5 / 1 (50%); 48% scored full marks. Mark distribution: 0: 52% · 1: 48%

Your mark: ____ **/ 1**

Question 9 (4 marks) · state average 2.3 / 4 (57%)

2022 VCAA Further Maths Exam 2 · Core · Question 7

a. Value of payment number 3

☐ $5214.28

Answer / notes: The payment doesn't change: $5214.28.

State average 0.9 / 1 (90%); 86% scored full marks. Mark distribution: 0: 14% · 1: 86%

Your mark: ☐ **/ 1**

b. Interest for payment 3, to the nearest cent

☐ $1330.65

Answer / notes: 0.25% of the balance after payment 2: 0.0025 × 532 261.78 = $1330.65.

State average 0.5 / 1 (50%); 51% scored full marks. Mark distribution: 0: 49% · 1: 51%

Your mark: ☐ **/ 1**

c. Recurrence relation for the balance P_n

☐ $P_0 = 540\,000$, $P_{n+1} = 1.0025P_n - 5214.28$

Answer / notes: 3% ÷ 12 = 0.25% a month. Common errors: the wrong factor, or adding the 5214.28.

State average 0.4 / 1 (40%); 35% scored full marks. Mark distribution: 0: 65% · 1: 35%

Your mark: ☐ **/ 1**

d. Monthly payment if the $540 000 were a simple perpetuity

☐ $1350

Answer / notes: A perpetuity pays only the interest: 0.0025 × 540 000 = $1350.

State average 0.5 / 1 (50%); 49% scored full marks. Mark distribution: 0: 51% · 1: 49%

Your mark: ☐ **/ 1**

Question 10 (4 marks) · state average 2.1 / 4 (53%)

2020 VCAA Further Maths Exam 2 · Matrices module · Question 3

a. State matrix one week after the sale

☐ $S_1 = (240\,700, 231\,700, 207\,600)$ for W, G, E

Answer / notes: $S_1 = TS_0 = (240\,700, 231\,700, 207\,600)$. Most errors were transcription slips.

State average 0.8 / 1 (80%); 79% scored full marks. Mark distribution: 0: 21% · 1: 79%

Your mark: ☐ **/ 1**

b. Change in Westmall's weekly shoppers from the sale to week 10, to the nearest thousand

☐ 30 000

Answer / notes: 250 000 at the start and about 220 000 at week 10 (read from the graph or calculated): a drop of 30 000.

State average 0.6 / 1 (60%); 64% scored full marks. Mark distribution: 0: 36% · 1: 64%

Your mark: ☐ **/ 1**

c. Week in which Grandmall reaches its maximum

☐ 7

Answer / notes: Calculate S_4 to S_{10}: Grandmall peaks at about 233 710 in week 7. Many guessed from the graph.

State average 0.3 / 1 (30%); 27% scored full marks. Mark distribution: 0: 73% · 1: 27%

Your mark: ☐ **/ 1**

d. Long-term weekly number of shoppers at Westmall, to the nearest whole number

☐ 218 884

Answer / notes: Raise T to a large power (e.g. S_{50}): Westmall settles at 218 884. 220 000 read from the graph wasn't accurate enough.

State average 0.4 / 1 (40%); 39% scored full marks. Mark distribution: 0: 61% · 1: 39%

Your mark: ☐ **/ 1**

Question 11 (2 marks) · state average 1.5 / 2 (75%)

2021 VCAA Further Maths Exam 2 · Matrices module · Question 1

a. Order of matrix J

☐ 3 × 1

Answer / notes: 3 × 1.

State average 1.0 / 1 (100%); 93% scored full marks. Mark distribution: 0: 7% · 1: 93%

Your mark: ☐ **/ 1**

b. Scalar k that increases every element by 5%

☐ 1.05

Answer / notes: k = 1.05. 0.05 would give only the increase.

State average 0.5 / 1 (50%); 48% scored full marks. Mark distribution: 0: 52% · 1: 48%

Your mark: ☐ **/ 1**

Question 12 (3 marks) · state average 2 / 3 (67%)

2021 VCAA Further Maths Exam 2 · Matrices module · Question 2

a. Two staff who can send information directly to each other

☐ Brie and Dex

Answer / notes: B → D and D → B are both 1.

State average 0.9 / 1 (90%); 92% scored full marks. Mark distribution: 0: 8% · 1: 92%

Your mark: ☐ **/ 1**

b. Sequence of links from Elena to Chai

☐ Elena – Dex – Brie – Chai

Answer / notes: E can only send to D, D only to B, and B to C.

State average 0.9 / 1 (90%); 83% scored full marks. Mark distribution: 0: 17% · 1: 83%

Your mark: ☐ **/ 1**

c. List both two-step links for the pair with two different two-step links

☐ Alex – Brie – Dex and Alex – Elena – Dex

Answer / notes: M^2 has a 2 in row A, column D: Alex reaches Dex through Brie and through Elena.

State average 0.2 / 1 (20%); 21% scored full marks. Mark distribution: 0: 79% · 1: 21%

Your mark: ☐ **/ 1**

Question 13 (2 marks) · state average 1.6 / 2 (80%)

2022 VCAA Further Maths Exam 2 · Matrices module · Question 1

a. Matrix equation for the cost of five nights of each type

☐ 5 × [80 140 270] = [400 700 1350]

Answer / notes: Multiply by the scalar 5: [400 700 1350].

State average 1.0 / 1 (100%); 94% scored full marks. Mark distribution: 0: 6% · 1: 94%

Your mark: ☐ **/ 1**

b. Column matrix A so that CA gives the cost of 2 motel nights and 3 apartment nights

☐ A = column (0, 2, 3)

Answer / notes: A = (0, 2, 3) as a column, in the order hostel, motel, apartment. Swapping the numbers around was common.

State average 0.6 / 1 (60%); 60% scored full marks. Mark distribution: 0: 40% · 1: 60%

Your mark: ☐ **/ 1**

Question 14 (3 marks) · state average 2.4 / 3 (80%)

2020 VCAA Further Maths Exam 2 · Networks module · Question 1

a. How many players Emerson had played with

☐ 1

Answer / notes: Emerson's vertex has degree 1.

State average 0.9 / 1 (90%); 95% scored full marks. Mark distribution: 0: 5% · 1: 95%

Your mark: ☐ **/ 1**

b. Who had played with both Alex and Bo

☐ Cameron and Dale

Answer / notes: Cameron and Dale are each joined to both. Many gave only one name.

State average 0.6 / 1 (60%); 58% scored full marks. Mark distribution: 0: 42% · 1: 58%

Your mark: ☐ **/ 1**

c. Add a new player, Finn, who had played with none of them

☐ An isolated vertex labelled Finn

Answer / notes: Draw a vertex for Finn with no edges.

State average 0.9 / 1 (90%); 91% scored full marks. Mark distribution: 0: 9% · 1: 91%

Your mark: ☐ **/ 1**

Question 15 (1 mark) · state average 0.6 / 1 (60%)

2020 VCAA Further Maths Exam 2 · Networks module · Question 2

Batting position for each player that maximises the team's score

☐ Alex 3, Bo 1, Cameron 2

Answer / notes: Check the six possible allocations: Alex 3, Bo 1, Cameron 2 gives 24 + 25 + 25 = 74, the highest. Some ran the Hungarian algorithm as if minimising.

State average 0.6 / 1 (60%); 57% scored full marks. Mark distribution: 0: 43% · 1: 57%

Your mark: ☐ **/ 1**

Question 16 (4 marks) · state average 3.4 / 4 (85%)

2021 VCAA Further Maths Exam 2 · Networks module · Question 1

a. Draw the missing edge

☐ The missing door edge drawn between the correct two vertices

Answer / notes: Check each door on the plan against the graph's edges and add the one that's missing.

State average 1.0 / 1 (100%); 93% scored full marks. Mark distribution: 0: 7% · 1: 93%

Your mark: ☐ **/ 1**

b. Degree of vertex E

☐ 2

Answer / notes: Room E has two doors.

State average 0.9 / 1 (90%); 88% scored full marks. Mark distribution: 0: 12% · 1: 88%

Your mark: ☐ **/ 1**

c.i. A route from F through every room once and back to F

☐ F–A–B–E–D–C–F (or the reverse)

Answer / notes: F–A–B–E–D–C–F, or the same in reverse.

State average 0.9 / 1 (90%); 92% scored full marks. Mark distribution: 0: 8% · 1: 92%

Your mark: ☐ **/ 1**

c.ii. Mathematical name for such a journey

☐ Hamiltonian cycle

Answer / notes: A Hamiltonian cycle (“cycle” alone wasn’t accepted).

State average 0.6 / 1 (60%); 60% scored full marks. Mark distribution: 0: 40% · 1: 60%

Your mark: ☐ **/ 1**

Question 17 (2 marks) · state average 1.6 / 2 (80%)

2021 VCAA Further Maths Exam 2 · Networks module · Question 2

a. Shortest distance from G to M

☐ 86 km

Answer / notes: G–O–N–M = 28 + 42 + 16 = 86 km.

State average 0.9 / 1 (90%); 85% scored full marks. Mark distribution: 0: 15% · 1: 85%

Your mark: ☐ **/ 1**

b. Town passed through twice on the shortest route that visits every town

☐ K

Answer / notes: L is joined only to K and M, and I only to H and K, so the route has to go through K twice, e.g. G–H–I–K–J–O–N–K–L–M.

State average 0.7 / 1 (70%); 65% scored full marks. Mark distribution: 0: 35% · 1: 65%

Your mark: ☐ **/ 1**

Question 18 (3 marks) · state average 2.3 / 3 (77%)

2022 VCAA Further Maths Exam 2 · Networks module · Question 1

a. A route that visits every cabin once and returns to the office

☐ O–A–B–C–D–E–F–G–O (or the reverse)

Answer / notes: Office–A–B–C–D–E–F–G–office, or the reverse. Give the whole route, starting and finishing at the office.

State average 0.8 / 1 (80%); 76% scored full marks. Mark distribution: 0: 24% · 1: 76%

Your mark: ☐ **/ 1**

b.i. Vertex where a route along every road exactly once, starting at the office, finishes

☐ E

Answer / notes: The office and E are the only odd-degree vertices, so the trail ends at E.

State average 0.8 / 1 (80%); 82% scored full marks. Mark distribution: 0: 18% · 1: 82%

Your mark: ☐ **/ 1**

b.ii. Name for that route

☐ Eulerian trail

Answer / notes: An Eulerian trail (not just “trail”, and not a circuit).

State average 0.7 / 1 (70%); 70% scored full marks. Mark distribution: 0: 30% · 1: 70%

Your mark: ☐ **/ 1**

Exam 1: ____ **/ 40** **Exam 2:** ____ **/ 61** **Total:** ____ **/ 101**

State average on these exact questions: Exam 1 32.7 / 40, Exam 2 41.3 / 61, total 74.0 / 101 (73%). The notes paraphrase the VCAA examiners' reports; read the full reports for sample answers.